Mixed-Strategy Equilibrium

Example: Matching Pennies

		Player 2	
		Head	Tail
Player 1	Head	1,-1	-1,1
	Tail	-1,1	1,-1

- Matching pennies does not have a Nash equilibrium (in the game with ordinal preferences), i.e. there is no steady state where all players 1 choose the same action and all players 2 choose the same action and nobody wants to change

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- But: we can assume that if a player faces getting 1 with probability p and -1 with probability $1-p$, then the higher p the better

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- Matching pennies does not have a Nash equilibrium (in the game with ordinal preferences), i.e. there is no steady state where all players 1 choose the same action and all players 2 choose the same action and nobody wants to change
- But: we can assume that if a player faces getting 1 with probability *p* and -1 with probability 1-*p*, then the higher *p* the better
- Now if player 2 chooses H with probability ½, then both H and T yield the same lottery to player 1 (namely getting 1 with probability ½ and -1 with ½)

Example: Matching Pennies

- Thus no matter which strategy player 1 chooses, she has no incentive to change it
- The same holds for player 2: if player 1 chooses H with probability ½ and T with probability ½, both H and T yield the same lottery for him
- Thus player 2 has no incentive to change his strategy
- Therefore, if both players choose H with probability ½ and T with probability ½, no one has an incentive to change
- Thus we have a steady state
- This can be interpreted either as a state where each player randomizes with equal probability between H and T or as game between populations where half of the players choose H and the other half choose T

Generalization: Preliminaries

- Ordinal payoff functions are generally not sufficient in order to express preferences over lotteries
- For example in this case we need to know how much worse is -2 than -1 compared to the difference of 1 and -1

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	Tail	-2,2	1,-1

- Thus we need a **cardinal** payoff function, where differences have meaning
- **von Neumann – Morgenstern (vNM) preferences:** Preferences over lotteries represented by the expected value of a payoff function over the deterministic outcomes (“Bernoulli payoff function”)

Definitions

- A **strategic game** (with vNM preferences) consists of
  - a set of players
  - for each player, a set of actions
  - for each player, **preferences** regarding lotteries over action profiles that may be represented by the expected value of a ("Bernoulli") payoff function over action profiles
- **Definition:** A **mixed strategy** of a player is a probability distribution over the player's actions
- can denote this by vector of probabilities $(p_1,\ldots, p_n)$ if A_i is a set of n actions $\{a_1,\ldots, a_n\}$
- Let $\Delta(A_i)$ be the set of mixed strategies over A_i
- **Pure strategy:** mixed strategy that puts probability 1 on a single action, i.e. a deterministic action

Mixed-strategy Nash Equilibrium

- Let α^* be a mixed strategy profile in a strategic game with vNM preferences
- Then α^* is a **(mixed-strategy) Nash equilibrium** if for every player i and every mixed strategy $\alpha_i \in \Delta(A_i)$:

$$U_i(\alpha^*) \geq U_i(\alpha_i, \alpha_{-i}^*)$$

where $U_i(\alpha)$ is i's expected payoff given α

Proposition: every game with vNM preferences with each player having finitely many available actions has a (mixed-strategy) equilibrium

Note: this does not mean that there is no equilibrium if players have infinitely many actions. There just might be no equilibrium

Best Response Functions

- Best response functions are defined as for deterministic actions
- **Best response function** of i:
- $B_i(\alpha_{-i}) = \{\alpha_i: U_i(\alpha_i,\alpha_{-i}) \geq U_i(\alpha_i',\alpha_{-i})$ for all mixed strategies $\alpha_i' \in \Delta(A_i)\}$
- Set-valued, each member of $B_i(\alpha_{-i})$ is a best response to α_{-i}

Again, we get:

Proposition: α^* is a **Nash equilibrium** if and only if

$$\alpha_i^* \in B_i(\alpha_{-i}^*) \text{ for every } i$$

Best Response Functions for Matching Pennies

		Player 2	
		Head	Tail
Player 1	Head	1,-1	-1,1
	Tail	-1,1	1,-1

- *r :* Probability that player 1 chooses Head
- *q:* Probability that player 2 chooses Head

Best Response Functions for Matching Pennies

		Player 2	
		Head	Tail
Player 1	Head	1,-1	-1,1
	Tail	-1,1	1,-1

- *r:* Probability that 1 chooses Head
- *q:* Probability that 2 chooses Head
- $B_1(q) = 1$ if $q > 1/2$; $= 0$ if $q < 1/2$; $=[0,1]$ if $q = 1/2$
- $B_2(r) = 0$ if $r > 1/2$; $= 1$ if $r < 1/2$; $=[0,1]$ if $r = 1/2$

Best Response Functions for Matching Pennies

		Player 2	
		Head	Tail
Player 1	Head	1,-1	-1,1
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- $B_1(q) = 1$ if $q > 1/2$; $= 0$ if $q < 1/2$; $=[0,1]$ if $q = 1/2$
- $B_2(r) = 0$ if $r > 1/2$; $= 1$ if $r < 1/2$; $=[0,1]$ if $r = 1/2$
- Equilibrium: $(r, q) = (1/2,1/2)$

Visualization of previous slides: Solving matching pennies

Rule: player 1 wins, when faces are opposite

player 2 wins, when faces are same

		Player 2			Expected payoffs
		Head	**Tail**		
Player 1	**Head**	-1 , 1	1 , -1	r	$1-2q$
	Tail	1 , -1	-1 , 1	$1-r$	$2q-1$
		q	$1-q$		

- r: Probability that 1 chooses Head
- q: Probability that 2 chooses Head
- Player 1's expected payoffs (calculated based on the probability of remaining player`s actions)
  - If Player 1 chooses Head, $q.(-1)+(1-q).1=1-2q$
  - If Player 1 chooses Tail, $q.1-(1-q).1=2q-1$

Visualization of previous slides: Solving matching pennies

Player 2

		Head	Tail		Expected payoffs
Player 1	Head	-1 , 1	1 , -1	r	$1-2q$
	Tail	1 , -1	-1 , 1	$1-r$	$2q-1$
Expected payoffs		q	$1-q$		
		$2r-1$	$1-2r$		

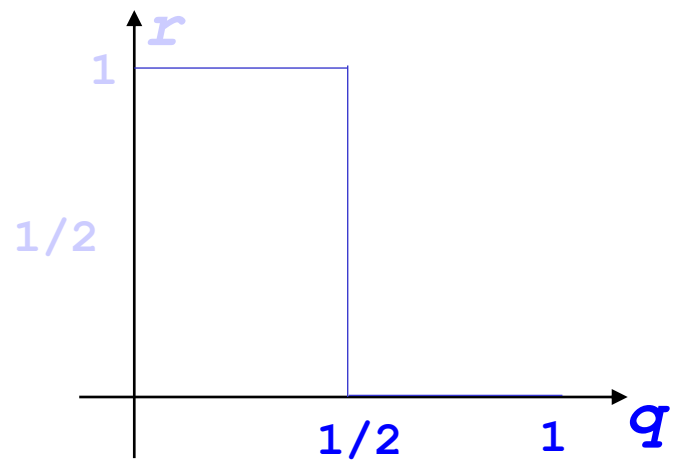
- r: Probability that 1 chooses Head
- q: Probability that 2 chooses Head
- Player 2's expected payoffs
  - If Player 2 chooses Head, $r.(1)+(-1).(1-r)=2r-1$
  - If Player 2 chooses Tail, $r.(-1)+(1-r).(1)=1-2r$

Visualization of previous slides: Solving matching pennies

Player 2

		Head	Tail	Expected payoffs
Player 1	r Head	-1 , 1	1 , -1	$1-2q$
	$1-r$ Tail	1 , -1	-1 , 1	$2q-1$
Expected payoffs		q $2r-1$	$1-q$ $1-2r$	

- Player 1's best response $B_1(q)$:
  - For $q<0.5$, Head ($r=1$)
  - For $q>0.5$, Tail ($r=0$)
  - For $q=0.5$, indifferent ($0 \le r \le 1$)

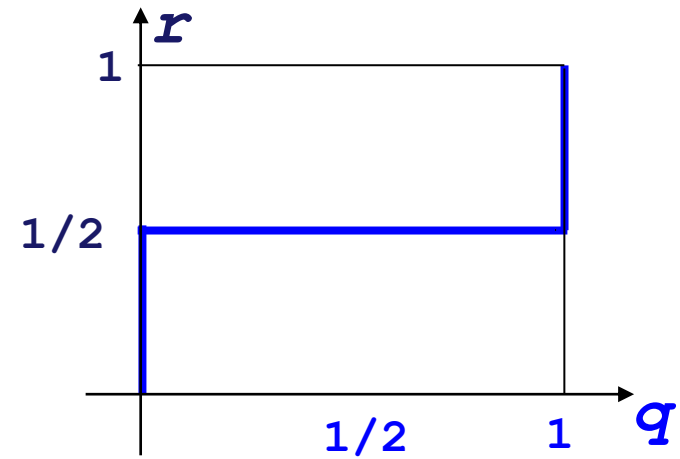


Note: In the Nash equilibrium, each player must be indifferent to choosing Head or Tail

Visualization of previous slides: Solving matching pennies

		Player 2: Head	Player 2: Tail	Expected payoffs
Player 1	Head	-1 , 1	1 , -1	r $1-2q$
	Tail	1 , -1	-1 , 1	$1-r$ $2q-1$
Expected payoffs		q $2r-1$	$1-q$ $1-2r$	

- Player 2's best response $B_2(r)$:
  - For r<0.5, Tail (q=0)
  - For r>0.5, Head (q=1)
  - For r=0.5, indifferent (0≤q≤1)



Note: In the Nash equilibrium, each player must be indifferent to choosing Head or Tail

Visualization of previous slides: Solving matching pennies

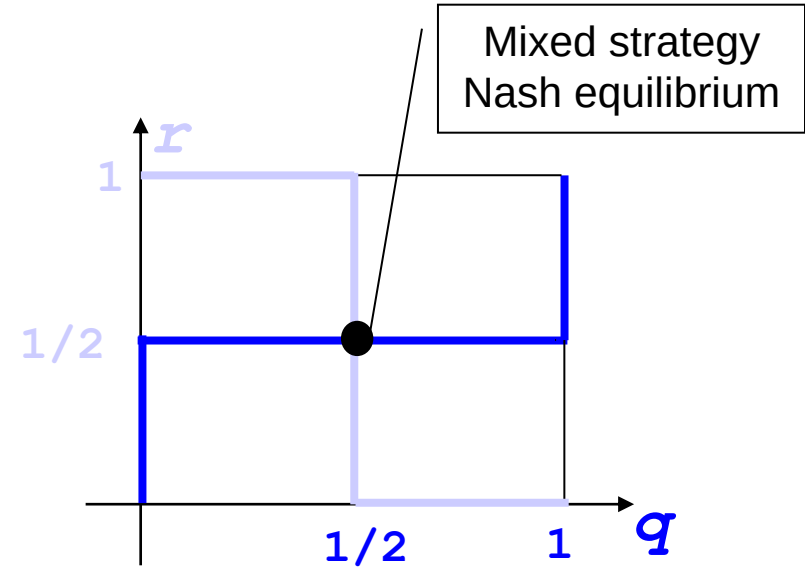
- Player 1's best response
  $B_1(q)$:
  - For $q<0.5$, Head ($r=1$)
  - For $q>0.5$, Tail ($r=0$)
  - For $q=0.5$, indifferent ($0 \le r \le 1$)
- Player 2's best response
  $B_2(r)$:
  - For $r<0.5$, Tail ($q=0$)
  - For $r>0.5$, Head ($q=1$)
  - For $r=0.5$, indifferent ($0 \le q \le 1$)

✓ Check

$r = 0.5 \in B_1(0.5)$

$q = 0.5 \in B_2(0.5)$

		Player 2		
		Head	Tail	
Player 1	Head	-1 , 1	1 , -1	r
	Tail	1 , -1	-1 , 1	$1-r$
		q	$1-q$	



Game 2

- Mix Strategy for Battle of Sexes Game

Best response function for Battle of the Sexes

		Player 2	
		Ball	Theatre
Player 1	Ball	2,1	0,0
	Theatre	0,0	1,2

- *p:* Probability that 1 chooses Ball
- *q:* Probability that 2 chooses Ball

Best response function for Battle of the Sexes

		Player 2	
		Ball	Theatre
Player 1	Ball	2,1	0,0
	Theatre	0,0	1,2

- *p:* Probability that 1 chooses Ball i.e. with (1-p) choses theatre
- *q:* Probability that 2 chooses Ball i.e. with (1-q) choses theatre
- $B_1(q) = 1$ if $q > 1/3$; $= 0$ if $q < 1/3$; $=[0,1]$ if $q = 1/3$
- $B_2(p) = 1$ if $p > 2/3$; $= 0$ if $p < 2/3$; $=[0,1]$ if $p = 2/3$

Best response function for Battle of the Sexes

		Player 2	
		Ball	Theatre
Player 1	Ball	2,1	0,0
	Theatre	0,0	1,2

- *p:* Probability that 1 chooses Ball
- *q:* Probability that 2 chooses Ball
- $B_1(q) = 1$ if $q > 1/3$; $= 0$ if $q < 1/3$; $=[0,1]$ if $q = 1/3$
- $B_2(p) = 1$ if $p > 2/3$; $= 0$ if $p < 2/3$; $=[0,1]$ if $p = 2/3$
- Equilibria: $(p, q) = (1,1) = (\text{Ball, Ball})$,

  $(p, q) = (0,0) = (\text{Theatre, Theatre})$

  $(p, q) = (2/3,1/3)$

Best response function

1. **Player 1's Best Response $B_1(q)$:**
   - If $q > \frac{1}{3}$: Player 1 chooses Ball with probability 1 (i.e., always chooses Ball).
   - If $q < \frac{1}{3}$: Player 1 chooses Ball with probability 0 (i.e., never chooses Ball, chooses Theatre instead).
   - If $q = \frac{1}{3}$: Player 1 is indifferent, so $B_1(q)$ could be any probability in the range $[0, 1]$.
2. **Player 2's Best Response $B_2(p)$:**
   - If $p > \frac{2}{3}$: Player 2 chooses Ball with probability 1 (i.e., always chooses Ball).
   - If $p < \frac{2}{3}$: Player 2 chooses Ball with probability 0 (i.e., never chooses Ball, chooses Theatre instead).
   - If $p = \frac{2}{3}$: Player 2 is indifferent, so $B_2(p)$ could be any probability in the range $[0, 1]$.

Pure-Strategy Equilibria survive allowing for mixing

- In battle of the sexes, the 2 equilibria identified in the game with ordinal preferences survive if we interpret the game as having vNM preferences and allow for mixing
- This is always true and vice versa for pure-strategy equilibria

In conclusion, The **Battle of the Sexes** game has:

- **Two pure-strategy Nash equilibria**: (Ball, Ball) and (Theatre, Theatre).
- **One mixed-strategy Nash equilibrium**: (p, q)=(2/3,1/3)
- In the pure-strategy equilibria, both players coordinate on the same activity, although each player has a different preferred outcome. In the mixed-strategy equilibrium, each player chooses their preferred activity with a probability that makes the other player indifferent between their options, resulting in a probabilistic mix of both activities.

Pure-Strategy Equilibria survive allowing for mixing

- **Proposition:** any equilibrium of a game with ordinal preference is a pure-strategy equilibrium of the corresponding game with vNM preferences and any pure-strategy equilibrium of a game with vNM preferences is an equilibrium of the corresponding game with ordinal preferences
- So all equilibria of the games we encountered so far stay equilibria if we allow for mixing and interpret the payoff functions as Bernoulli payoff functions

Characterization of mixed-strategy equilibria

Proposition: let α^* be a mixed-strategy equilibrium. Then

- each action a_i that is played by i with positive probability according to α_i^* yields the same expected payoff to i as strategy α_i^*
- every action a_i' that is played by i with probability 0 according to α_i^* yields at most the same expected payoff to i as strategy α_i^*

This is useful for finding mixed-strategy equilibria:

- each player has to be indifferent between all the strategies she plays
- so for a given game and each player i, we can consider subsets of strategies and see whether we find solutions for the **other players' mixed strategies**, such that i is indifferent among all actions in the subset

Symmetric games

Definitions of symmetric games and symmetric equilibria carry over in obvious way

We had seen that symmetric game with ordinal preferences does not necessarily have a symmetric equilibrium

But a symmetric game with vNM preferences and a finite set of actions **always** has at least one symmetric equilibrium (pure or mixed).